URSI GASS 2026, Krakow, Poland, 15 – 22 August 2026

Propagation Modeling over a Piecewise Terrain in 3D using the Shift Map Method including Cross Terms

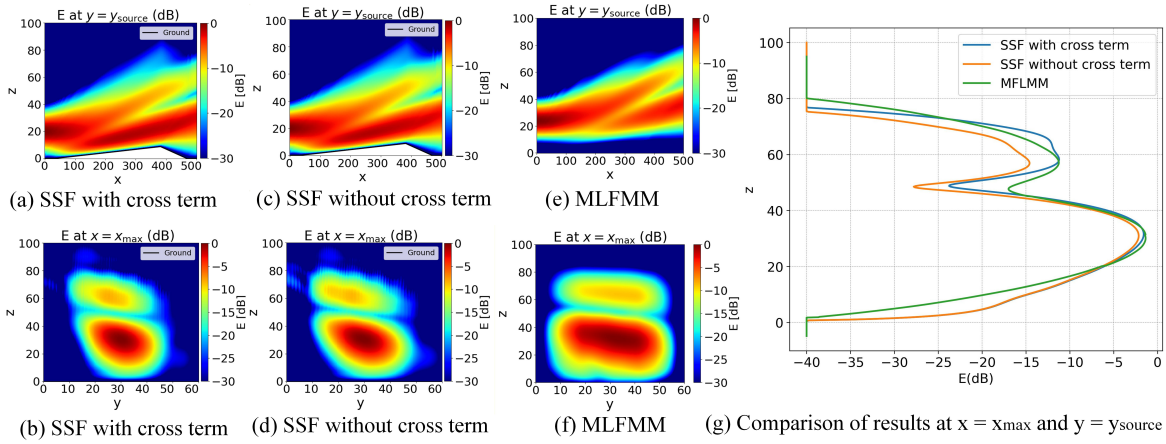
Xinyi Huang*(1), Alexandre Chabory(1), Rémi Douvenot(1), and Hang Zhou(2)
(1) ENAC University of Toulouse, France
(2) Civil Aviation University of China, Tianjin, China

Extended Abstract

The split-step Fourier (SSF) method provides an efficient numerical solution to the parabolic equation (PE) for modeling electromagnetic wave propagation. Shift map method[1] combined with SSF has been widely applied to wave propagation over terrain in two dimensions (2D), however, extending this approach to three dimensions (3D) becomes more challenging due to the increased complexity of handling terrain profiles. In 3D propagation problems, a cross term is introduced in the PE, which is commonly neglected for computational simplicity but impacts to the accuracy. In the previous works, the terrain profiles have often been approximated as wedge shaped[2] to simplify analysis by neglecting lateral effects. However, such a simplification may not be applicable in more complex conditions.

In this work, the terrain profile is defined as a piecewise surface with variations in both x and y directions, which is explicitly incorporated into the narrow-angle parabolic equation (NAPE) through a coordinate transformation based on the shift map method. Crucially, the cross term is kept to account for lateral coupling effects, thereby enhancing modeling accuracy. In the end, standard PE in transformed coordinate is retrieved where a phase corrector term is computed separately.

To validate the effectiveness of the proposed algorithm, a piecewise relief constructed as perfect conducting ground with pyramid like shape is applied to simulate the terrain profile. The simulated propagation is compared with the results with cross term neglected and the ones of MLFMM method using FEKO, as given by Fig. 1. It observed that the field intensity retrieved with the cross term is stronger than that without the cross term when $x > x_{peak}$. Moreover, the method including the cross term shows better agreement with the MLFMM at the final iteration step at $y = y_{source}$.



(a) SSF with cross term (c) SSF without cross term (e) MLFMM

(b) SSF with cross term (d) SSF without cross term (f) MLFMM (g) Comparison of results at x = x_{max} and y = y_{source}

Figure 1. Simulation results.

References

[1] D. J. Donohue and J. R. Kuttler, "Propagation modeling over terrain using the parabolic wave equation," *IEEE Transactions on Antennas and Propagation*, **48**, 2, 2002, pp. 260–277, doi:10.1109/8.833076.

[2] C. Khodr, M. Azarpeyvand, and D. N. Green, "An iterative three-dimensional parabolic equation solver for propagation above irregular boundaries," *The Journal of the Acoustical Society of America*, **148**, 2, 2020, pp. 1089–1100, doi:10.1121/10.0001766.